Scalability & Future Plan

Date	02 July 2026
Team ID	SWTID-2026-4638
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:
This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Limited support for integration with live banking and financial verification systems	Reduces the ability to perform realtime applicant verification and automated decision-making	High
2	Prediction history is stored locally without centralized database support	Restricts long-term data management, reporting, and scalability for large numbers of users	High
3	Model performance depends on the quality and completeness of applicant data	Inaccurate or missing information may affect prediction accuracy and decision reliability	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a single server	Deploy on cloud infrastructure (AWS, Google Cloud, Azure) with load balancing and auto-scaling capabilities
2	Data Storage	Prediction history is maintained using local JSON files	Implement scalable databases such as PostgreSQL or MongoDB with cloud storage integration

3	Performance	Machine learning predictions are generated in real time for each request	Introduce caching mechanisms, asynchronous processing, and optimized model inference pipelines
4	Security	Basic application-level validation and local data storage	Implement user authentication, encryption, role-based access control, and secure API communication

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integration with banking systems, credit bureaus, and financial verification services	2 Months	Enables automated applicant verification and improves prediction accuracy
Phase 3	Explainable AI, advanced risk analysis, and automated model retraining	4 Months	Improves decision transparency, model performance, and user confidence
Phase 4	Cloud deployment, analytics dashboard, mobile application, and multilingual support	6 Months	Enhances scalability, accessibility, user experience, and enterprise adoption